A single predictive model shut down an entire business line of a major company.

Zillow Offers · iBuying · November 2021

$500M+

total losses — an entire business line shut down

$300M+	~2000	−25%
write-down for the quarter	laid off (≈25%)	stock in days

The model forecast the price of a house · Zillow automatically bought, renovated, and resold thousands of houses · systematically overvalued them

This was not ChatGPT

A predictive model estimated a number (a price) from tabular and geo-data. An LLM was not — and could not be — used here: the task is not textual.

What type of AI was this — and why did an ordinary model error become a business loss, rather than a minor inaccuracy?

A forecast error is normal. The business collapsed because of WHERE it was wired: to an automated, irreversible action at the scale of thousands of houses.

LECTURE 5

AI in Finance and Retail

05

> For which task — which type of AI, why that one, and where it breaks

One type of AI in depth (L4) → a palette of types for a palette of tasks (L5).

L2: the «which AI» tree — ML / deep learning / LLM → **L5: five structurally different types for different tasks**

L4: one type (LLM coder) in depth along autonomy → **L5: a palette of types for a palette of tasks in one industry**

Map of five AI types for five different tasks (+ a cross-cutting CV layer):

- **Time-series forecasting**
- **Anomaly detection**
- **ML scoring**
- **LLM assistants**
- **Recommendations and pricing**

Most of the value here comes NOT from a language model. The LLM is not a universal hammer: a different task requires a different type of AI. That is exactly «a palette of types for a palette of tasks».

The central question of the lecture.

«Finance and retail are the industries with the highest AI adoption. For which task — which type of AI, why that one (and not LLM everywhere), and where does this type break?»

We examine each type by one scheme (5 steps):

1. task → 2. type of AI and why it → 3. real example → 4. where it breaks → 5. alternative and criterion

The map is the structure of the lecture, not a requirement to understand everything now. Forecasting · anomalies · scoring · LLM · recommendations (+ a cross-cutting CV).

The answer is not «AI is good» and not «AI is bad», but an apparatus: name the appropriate type, justify the choice, and see in advance the point where a human is mandatory. By the end this scheme will become your tool of choice.

SECTION 1

Forecasting: demand, sales, churn

We start with retail's most widespread task — and right away with a type of AI that is NOT an LLM.

Time-series forecasting and a language model solve structurally different tasks.

Language model (LLM)	Time-series forecasting
predicts the next TOKEN of text	predicts the next NUMBER in the series
optimized for the plausibility of the wording	needs numeric accuracy + calibrated uncertainty
has no internal notion of «seasonality» / «trend»	series = trend + seasonality + noise
has seen texts ABOUT sales	sees the sales series itself as a series

Type of AI for forecasting: classical statistics and tabular ML — the ARIMA family, gradient boosting. NOT a generative model, NOT an LLM.

Hand «forecast the demand» to a text model — it will produce a plausible number with no connection to seasonality: a hallucination in a numeric wrapper. The type of AI is determined by the STRUCTURE of the task, not by the trendiness of the tool.

Why a tabular model here, not an LLM — three arguments, not a caveat.

1. Data structure

tabular numeric series (date, point, product, price, features). Series models extract trend and seasonality; turning a series into text for an LLM loses exactly this structure.

2. Measurability and calibration

the decision about order size is built on a numeric error metric + a confidence interval. Classical models give this out of the box; an LLM does not.

3. What will break

an LLM will give a plausible number with no grounded uncertainty — and the decision still has to be made, and the cost of a systematic error × volume (the Zillow mechanism).

Alternative and criterion: the right tool is a tabular predictive model with a measurable error. Even it is not enough under a non-stationary environment + an irreversible capital action without human control.

Tasks: demand forecasting (shelf / procurement), cash-flow (liquidity), customer churn (whom to retain).

If the series contain PII of Russian citizens, sending them to a public cloud LLM violates localization requirements (Federal Law 152-FZ); a tabular model is deployed on-premise.

The forecast is dictated by the task: neither X5 nor Magnit chose an LLM.

X5 Group · «Pyaterochka», «Perekrestok»

Demand forecast accuracy	> 70%
Extra revenue from ML tools	+5 bn ₽
Expiry write-offs	–2%

Magnit — import substitution of forecasting systems

Before 2022: foreign vendors of the SAP / Blue Yonder class

After they left: own forecasting + auto-ordering system (F&R)

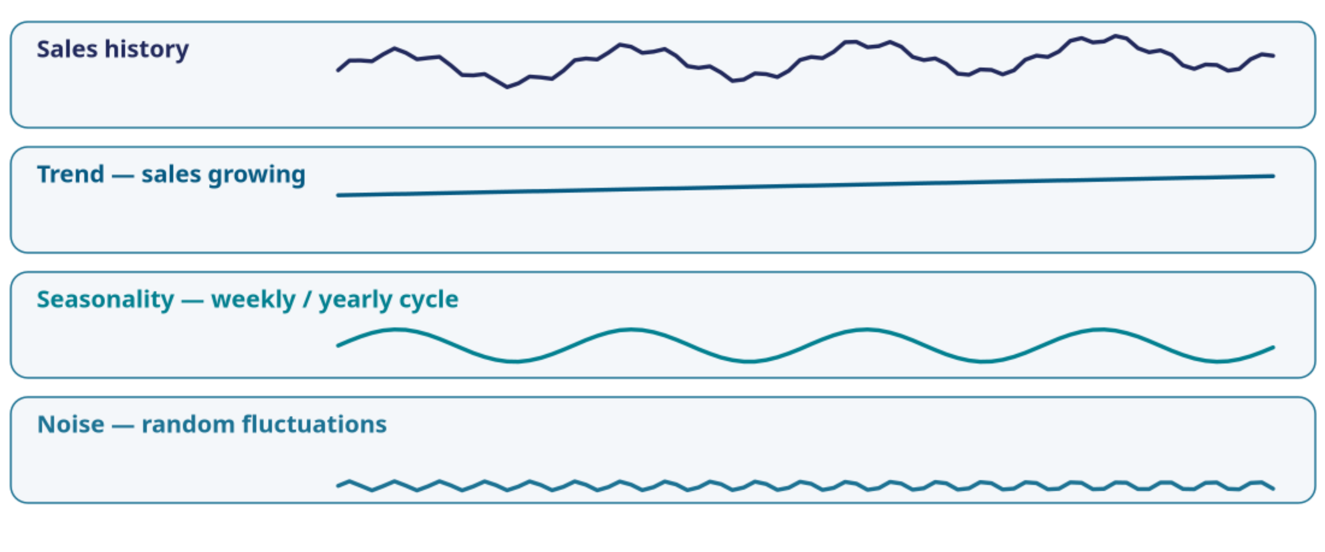
Status: pilot at a distribution center (2024–2025)

The departure of foreign vendors made import substitution of forecasting systems a direct engineering task for the industry.

Neither X5 nor Magnit solves the task with a language model — they build specialized forecasting systems, because the TYPE OF AI IS DICTATED BY THE TASK.

Per company data and industry reviews (X5/TAdviser 2023; shoppers.media/TAdviser 2024–2025); the numbers are as claimed by the companies, verified on the day of the lecture.

A forecast extends the patterns of the past — here lie both its strength and its vulnerability.



Any sales series = trend + seasonality + noise; the model extends the regular part.

What the model does — numerically

- decomposes history into trend + seasonality + noise
- learns the regular patterns and extends them
- adds corrections for promo / holiday

Forecast → action: «demand ≈ N» → «order N + buffer»

Analogy: a forecasting model is like an experienced buyer «stock up more for the weekend», but for millions of store × product pairs at once.

Built-in vulnerability: a forecast is strong exactly as long as TOMORROW RESEMBLES YESTERDAY. When the environment changes qualitatively — the model confidently extrapolates patterns that no longer exist. The key to the next slide.

What ruined Zillow was not model drift, but what its output was wired to.

distribution shift

the real data stopped resembling the training data — the housing market of 2020–2021 became statistically different

asymmetry of the cost of error

a recommendation error ≈ 0 · the same error on which a house is bought = tens of thousands of $ × N, and IRREVERSIBLE

Three things together = fatal

irreversible action × automated on the model's output × a non-stationary environment without a circuit-breaker (auto-stop on anomaly)

Knight Capital, 2012: $440M / ~45 min on a deterministic algorithm — the same class of error

Zillow vs Opendoor

The same type of AI, the same period. Opendoor survived thanks to a more conservative spread and risk design.

It is dangerous when simultaneously: (a) the output triggers a large irreversible action automatically, (b) the environment is non-stationary, (c) there is no circuit-breaker.

Alternative: a narrow segment + human-gate + live error-monitoring + circuit-breaker.

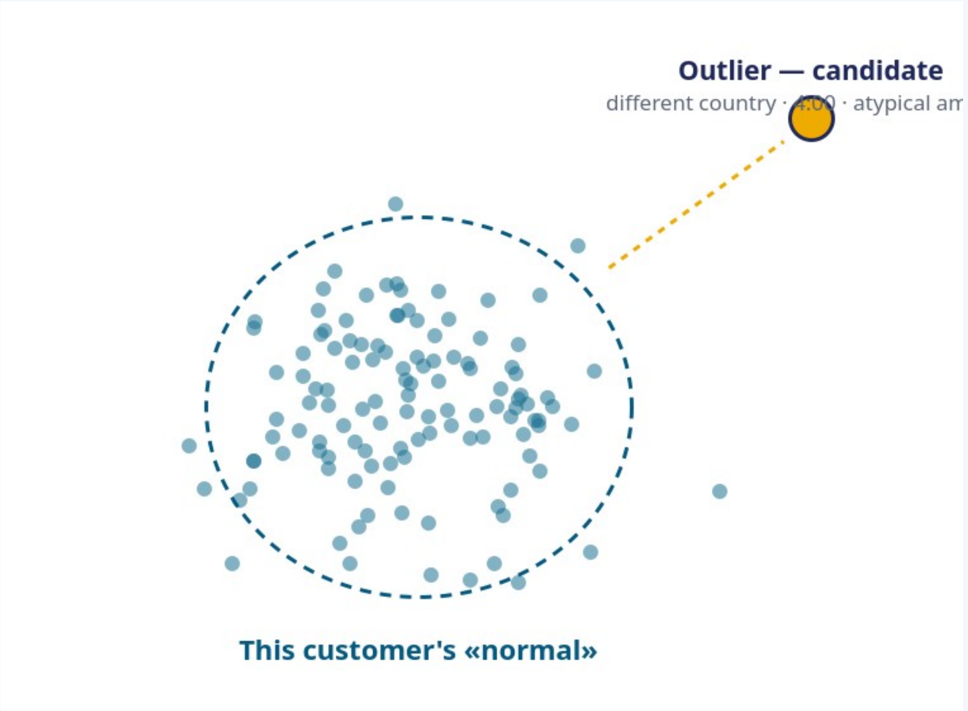
The model did exactly what it was meant for. The error was the engineering decision of WHERE to wire its output. The same AI, a different judgment → bankruptcy vs survival.

SECTION 2

Anomaly detection: fraud and AML

Forecasting is about the future; now about the present — catch an anomaly within milliseconds, while the payment has not yet gone through.

Learn the «customer's norm» and catch the deviation — not learn «what fraud looks like».



Task: catch a fraudulent transaction in real time, where almost everything is legitimate and the share of fraud is extremely small.

Type of AI — anomaly detection: the model builds the customer's «norm» and signals deviations.

not classification	fraud is rare and constantly changes its form (a moving target)
not time-series forecasting	the question is not «the next value», but «how unlike the norm»
not an LLM	the transaction is structured; the task is geometric, not linguistic

AML is a subset of the same task; hard statutory thresholds = deterministic rules, not a model.

The model does not «know» that this is fraud — it knows that this is NOT LIKE THE NORM, and raises a flag for review.

Anti-fraud is «norm / deviation» in real time, not generation.

What anomaly detection delivers in anti-fraud — each metric in its own units:

Stripe Radar

−32% fraud

while approving legitimate > 99%

JPMorgan

−30% false

fewer false positives

Visa · prevented

~$40 bn

of fraudulent transactions (FY2023)

Russia

Anomaly detection is standard practice at major banks (per Bank of Russia materials, 20.11.2025).

Notice: «reduction of FALSE positives» — the key metric of anti-fraud is NOT «accuracy in general», but the ratio of the two types of error. We will examine this further.

Different units — NOT one scale: % reduction, % false, bn $. And still the main metric is not «accuracy», but FP / FN separately.

Stripe/JPMorgan/Visa — as claimed by the companies; Visa — per Reuters/CNBC reports, July 2024; Russia — per Bank of Russia materials, 20.11.2025; the numbers are verified on the day of the lecture.

«99.9% accuracy» under strong imbalance is deceptive.

	Was fraud	Was legitimate
System: fraud	**TP** caught (good)	**FP** honest customer blocked (type I error)
System: pass	**FN** money went to the fraudster (type II error)	**TN** correctly passed (good)

Why accuracy lies

A stream of 1,000,000, fraud 1000 (0.1%). A model that says «everything is legitimate» gives 99.9% accuracy — and is completely useless.

cost-sensitive

FP and FN differ in COST (money, trust). Minimize the total expected cost, not a «pretty» accuracy.

The formal apparatus (sensitivity / specificity) is built in Lecture 7 on a medical example; here — the working intuition.

The first question to any anti-fraud number is NOT «what is the accuracy», but «what are the FP and FN separately, and at what cost».

«We cut FP by 25%» averages a penny FP and a catastrophic FP.

The same class of error — cost differs by orders of magnitude

FP #1 · $5 for coffee

the customer retried, forgot within an hour → cost ≈ 0, REVERSIBLE

FP #2 · $5000 for treatment abroad

hard to get through, time is critical → IRREVERSIBLE in its consequences

precision ↔ recall — a TRADE-OFF, not a task of «both to zero». A better model shifts the curve; the point is chosen by the engineer per the cost of error.

Auto-block of the irreversible without a gate — the Zillow / Knight class

Criterion:

- auto-hard block — only for the reversible / small
- large / irreversible → a soft challenge (3DS, call, push) + a fast human channel for unblocking
- a hard AML threshold — a rules-engine (law, not «with probability 0.97»)

The cost of FP averages the penny and the catastrophic — measure them separately.

Scale does not reduce harm: 0.5% false of billions = tens of millions of blocked legitimate transactions. Behind each one — a specific person.

SECTION 3

Credit scoring: why NOT a neural network

The anomaly has been found; now a decision that changes the customer's life — and why a black box is not allowed here.

3

A black box in scoring is inapplicable in principle — not «less convenient».

1. Explainability = law

a customer who is declined is legally owed an explanation (reason codes: «high debt burden»). «The algorithm decided so» is impermissible.

2. The data is tabular

deep networks are strong on text/audio/image; on tabular data, boosting is competitive at incomparably greater interpretability.

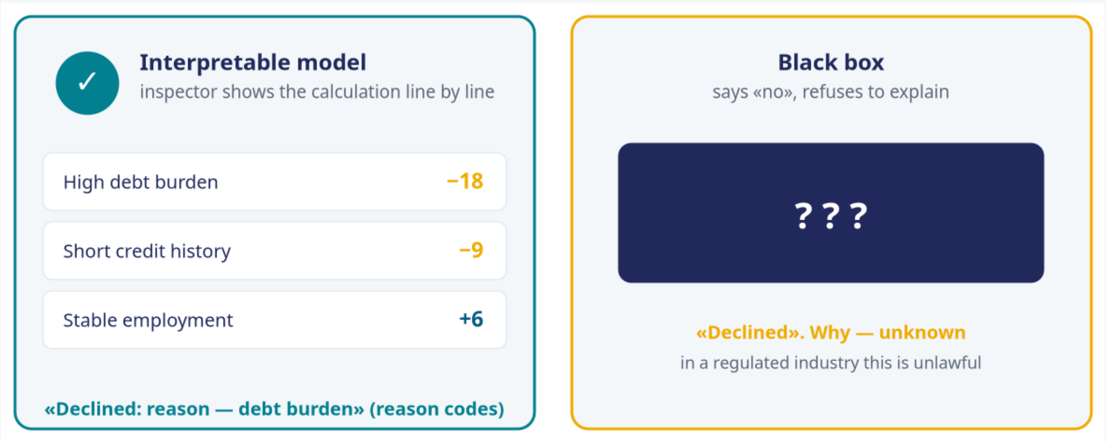
3. Audit > +1% accuracy

the regulator must reproduce the decision, verify the absence of discrimination. A stable, explainable one matters more than an opaque, more accurate one.

4. What breaks with a black box

cannot be explained → a violation · cannot be audited for bias → a crisis (Apple Card) · unstable → cannot be defended before the regulator.

Anchor analogy: a credit inspector



An interpretable model shows the calculation line by line (reason codes); a black box refuses to explain.

«New = a neural network, so better» in scoring is a structural error: explainability here is not a bonus, but a CONDITION OF LEGALITY.

«100% AI decisions» in Russia ≠ a neural-network black box out of control.

Sberbank · since February–March 2024

Decisions on individuals

~100% AI

Scoring model

up to 5000 parameters

AI effect across all lines, 2023

+350 bn ₽

«100% AI» actually means:

- high automation of the pipeline on interpretable models
- reason codes on every decision
- regulatory supervision by the Bank of Russia
- **the customer's retained right to a human: > 80% of financial organizations offer a human opt-out**

The type of AI — tabular ML, chosen because of explainability, not in spite of it. The lecture's thesis at the scale of the largest bank.

Per the bank's statements (TAdviser/AdIndex/Interfax, 2024) and Bank of Russia materials (20.11.2025); the numbers are verified on the day of the lecture.

Even a formally lawful model without explainability creates a crisis.

Apple Card / Goldman Sachs · 2019–2021

- Nov 2019: a viral thread — the limit is ×20 higher for the husband (while the wife's credit rating is higher)
- NYDFS opened an investigation (~400,000 applicants)
- **March 2021: NYDFS found NO violation of the law**
- BUT: customers received no explanations → a regulatory-reputational crisis

To claim «discrimination was proven» is factually incorrect.

Proxy bias «in plain terms»

The engineer does not feed in gender/race. But features remain that CORRELATE with them: postal code, spending history, employment. The model, optimizing accuracy, INDIRECTLY reconstructs the forbidden attribute through a proxy — without any intent.

Criterion: a regulated decision requires simultaneously reason codes + an appeal path (a human on the contested one) + a bias audit of outcomes BEFORE production.

The canonical analysis of the mechanism (Obermeyer/Optum) — Lecture 7.

«We do not use protected attributes» is a NECESSARY but wholly insufficient condition. The only proof is a direct audit of outcomes by group.

The higher the autonomy, the stricter the gate around it must be.

One class of error — three different types of AI:

Zillow
forecast → automated irreversible buying spree

fraud auto-block
anomaly → automated irreversible block

Knight Capital
deterministic algorithm → automated irreversible orders

Automation executing irreversible financial actions in an open loop without a kill switch (a manual «stop everything»), without limits on volume/position, without a circuit-breaker (auto-stop on anomaly), and without verified deployment, turns an ordinary model error into the SPEED OF RUIN.

Scoring in Russia is ~100% automated → the criterion applies acutely. «100% AI» is acceptable ONLY within a harness: reason codes + a human channel + a bias audit + drift monitoring + supervision. Remove the harness — and it is Apple Card at the scale of a bank.

High autonomy is neither the goal nor evil in itself. The goal is autonomy surrounded by paid-for gates, proportional to the cost of error and the irreversibility of the action.

SECTION 4

LLMs in finance: assistants and fact-checking

Until now — deliberately NOT an LLM. Now the LLM — and right away about its limits in a regulated industry.

An LLM is good exactly where the task is textual-conversational — and bad where it is not.

Where an LLM is the right type in a bank:

- customer support — answers to typical questions
- voice assistant — recognize the request, guide through the flow
- help for an employee — summarize an inquiry, suggest the regulation
- explaining a product in plain language

This is exactly what an LLM is designed for (unlike Sections 1–3).

Scale (verified): T-Bank's chatbot > 40% of inquiries; ~70% of banks planned voice by 2025.

Why NOT «> 90% of banks' inquiries»

The ~90% figure is real — but it is the share of calls handled by a voice assistant IN ONE BANK'S CALL CENTER, not the share of banks' inquiries overall. This is BASE SUBSTITUTION (fact-checking class 5).

The lecture teaches fact-checking — building an assertion on a figure with a substituted base would violate exactly the principle it teaches. A precise modest figure is worth more than a loud one.

Symmetry with Sections 1–3: «different task — different type of AI» means not «LLM is bad», but «an LLM is good exactly where the task is textual-conversational, and bad where it is not».

Here an LLM is not a source of truth, but an interface to a source of truth.

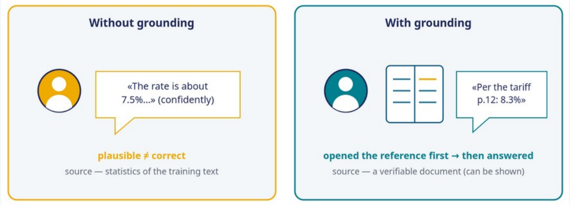
When an LLM is NOT a source of truth — 3 arguments

mechanism	generates the plausible, does not retrieve a fact («plausible» ≠ «correct»)
cost	a wrong rate/term = a violation, the organization is liable
what will break	free generation about facts = a generator of disinformation with legal liability

Five classes of error in an AI claim

1. hallucinated fact → check against the primary source
2. outdated data → check the date/currency
3. proxy bias in the output → audit outcomes, not inputs
4. deception by metric → FP/FN separately in money
5. base substitution → ask «share of what out of what»

Grounding anchor analogy: a student guesses in a confident tone vs first opens the reference book



Alternative: a fixed fact → deterministic retrieval (grounded RAG), not generation. In the lecture — recognize the class (Understand); verify 5 claims against primary sources — Seminar 5 (Apply).

The right type of AI without grounding and a human exit = Air Canada / Klarna.

Case E — Air Canada (callback)

- the chatbot stated a nonexistent refund policy
- the passenger acted on the answer
- Feb 2024: the court — the company IS liable for the info of its chatbot, ordered to pay compensation
- class: hallucinated financial fact = legal liability

Case F — Klarna (the 2023 → 2025 arc)

- LLM assistant: ~2/3 of inquiries, ~11 min → < 2 min
- claimed savings ~$40M/year (Klarna, 2024)
- 2024: framed as «AI replaces support»
- mid-2025: CSAT↓ → returned to hiring people

Criterion: a fixed policy/tariff/right → deterministic retrieval/grounded, not free generation; and in any scenario a path to a human is guaranteed.

What is sustainable is not replacement, but AUGMENTATION: AI on the mass routine + guaranteed human escalation on the emotional / disputed / non-standard tail.

Checkpoint: 4 of 5 types covered — a turn toward «human + harness».

Progress across the 5 types — for all 4 covered, a failure of one form:

- ✓ **Forecast** — Zillow
- ✓ **Anomalies** — fraud-FP
- ✓ **Scoring** — Apple Card
- ✓ **LLM** — Air Canada
- → **Recsys** — next

Level 1 «which type of AI» — resolved by the structure of the task, often unambiguously. Level 2 «what surrounds the AI at the cost of error» — is designed separately. The lecture's turn: from here on, attention moves from the type to the harness.

The last type — recsys / pricing — seems the most HARMLESS: the cost of error is near zero. That is exactly why it is more dangerous than the rest: the failure is SILENT — the system reports success by its own metric.

The question for the remaining section is NOT «which type», but «why can the type most harmless by cost of error turn out to be more dangerous than the loud failures?»

SECTION 5

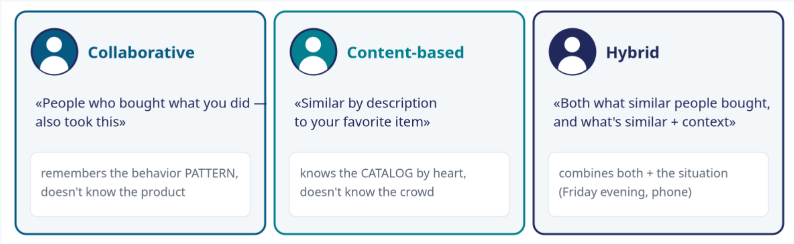
Recommendations and dynamic pricing

The last type — the most «harmless» by cost of error, but with the subtlest pathologies.

Two basic approaches to recommendations — and each with a weakness by name.

Collaborative — «ask those similar to you»	Content-based — «more like it, by description»
user × item matrix	product attributes (genre, brand)
⊕ catches unexpected connections	⊕ no cold-start for a new product
⊖ cold-start + popularity bias	⊖ over-specialization (niche)

Anchor analogy: three sellers (collaborative · content · hybrid)



Remember: WHAT these approaches are and WHICH weakness by name each has — collaborative (behavior pattern) vs content (the essence of the product).

A hybrid softens the weaknesses — but without explicit diversity it does not cancel the bubble.

Hybrid recommender

collaborative + content-based + context. The content part closes cold-start, the collaborative one breaks over-specialization. Most production systems.

Amazon	Netflix
~35%	**~75%**
of revenue — from recommendations	of viewing — from recommendations

A historical estimate (dating back to McKinsey ~2013), NOT a fresh verified figure — hence a number, not a chart-«fact».

Filter bubble

showed something similar → the user interacts with the similar → the model shows something even more similar. The narrowing of diversity is NOT malicious intent, but a consequence of optimizing for short-term relevance.

Dynamic pricing

auto-adjustment of price to demand/time/context. The perception of fairness and the law are CONSTRAINTS of the task, NOT variables to optimize.

The filter bubble arises without any intent — a direct consequence of «guess what you'll buy now». A hybrid does NOT automatically cancel this narrowing.

Amazon ~35% / Netflix ~75% — a historical estimate (McKinsey ~2013), NOT a fresh figure; Ozon/WB — the companies do not disclose the exact share.

The system reports success by its own metric — while destroying what did not make it into the metric.

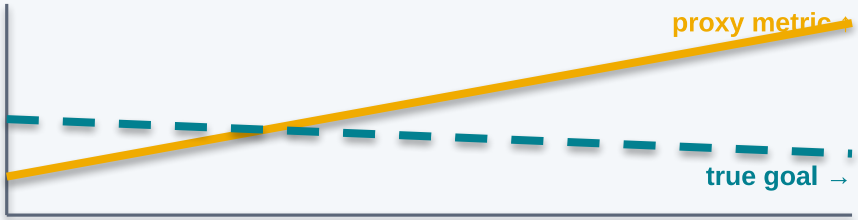
The root — proxy ≠ goal

the system optimizes what is measurable in the moment (the PROXY: clicks, time, margin) ≠ the real goal (trust, well-being). The divergence → filter bubble, dark patterns, homogenization.

Case G — Wendy's, February 2024

~$20M into digital menu boards with dynamic pricing → the media read it as surge-pricing → #BoycottWendys, a competitor outplayed it in advertising → rollback within days. The money was invested, the technology was available — what failed was ignoring that perceived fairness = a CONSTRAINT, not a variable.

Why the failure is SILENT:



dashboard green, trust falling

Criterion:
- proxy ≠ goal → serendipity + explainability + an audit for discrimination
- the cost is a decision of a HUMAN IN A LEGAL FRAME, not the output of the optimizer

One class runs through the whole lecture: accuracy deception, proxy bias in scoring, Klarna, filter bubble — PROXY ≠ GOAL. The more powerful the optimizer, the costlier the divergence. The failure is silent.

Name the type for the structure of the task — and sometimes the right type is not AI.

Task	Type of AI · why that one	Typical failure
Demand forecasting / cash-flow / churn	Time-series forecasting (ARIMA/boosting) — numbers with a trend, not text	distribution shift × irreversible automation (Zillow)
Real-time fraud / AML	Anomaly detection + rules-engine — norm/deviation	FP at scale; accuracy lies under imbalance
Credit scoring	Tabular ML (logreg/GBM+SHAP) — explainability = law	proxy bias + opacity (Apple Card)
Support / voice / explanation	LLM (grounded) — a textual-conversational task	hallucinated financial fact = legal liability (Air Canada)
Recommendations / pricing	Recsys; pricing — an optimizer within a frame	proxy≠goal: filter bubble (Wendy's)
A deterministic regulatory task (a hard AML threshold)	**Ordinary code / rules-engine — NOT AI; law ≠ probability**	AI would add non-determinism + an error surface

The bottom row — deliberately. A hard, verifiable regulatory rule → deterministic code, NOT AI: AI would add non-determinism where accuracy and auditability are needed.

The substantive answer to the central question of the lecture.

In none of the five failures is the fix «a better model». In all of them — a better JUDGMENT about what stands around the AI at the cost of error.

Five cases from the lecture — one and the same remedy (NOT a model):

Zillow	↓	human-gate
Apple Card	↓	explainability + audit
Air Canada	↓	grounding
Klarna	↓	augmentation
Wendy's	↓	legal frame

This is the answer: not «AI is good/bad», but «name the type, justify the choice, and design the harness for the cost of error».

Financial data, PII (personal data), and biometrics must not go into a public LLM.

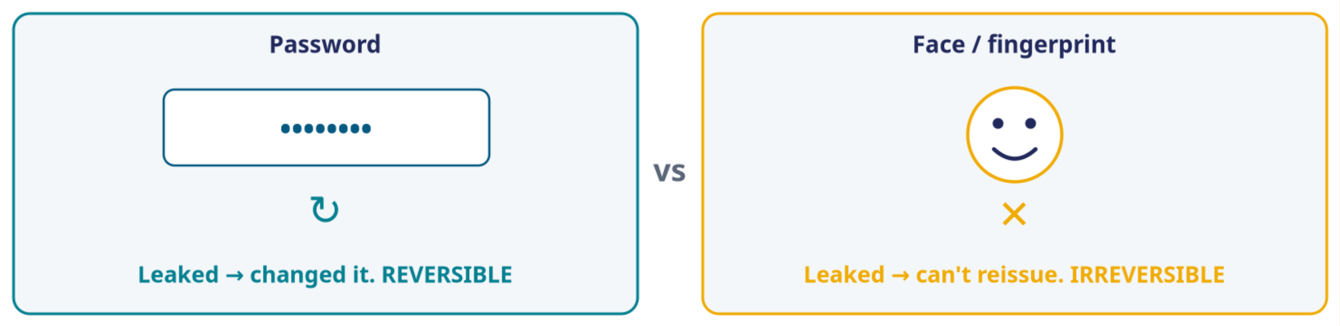
(A) Data and the law: 152-FZ / PII

- financial data + PII + biometrics = sensitive; Federal Law 152-FZ — localization of personal data of Russian citizens, biometrics = a strict regime
- public cloud: the data leaves the loop · no control of retention · auditability drops
- the criterion is by data sensitivity and regime, NOT by the strength of the model

(B) The CV layer: KYC, biometrics, hidden labor

- KYC = identifying the customer; liveness — a live human in front of the camera, not a photo/mask/deepfake (a strict regime)
- Just Walk Out (Amazon) shut down in 2024: the «autonomous» checkout relied on > 1000 reviewers in India — hidden labor
- Computer-vision bias is deepened in Lecture 7.

Biometrics are irreversibly compromised on a leak — the same logic «irreversible → a strict gate» that ran through the whole lecture.



«Fully autonomous» in marketing is a hypothesis to be tested, not a fact (Amazon disputed the scale).

Before applying AI to a financial / retail task:

1. Which type of AI does the structure of the task dictate?
2. Why NOT an LLM here? (≥1 structural reason)
3. Can it be solved without AI at all? (rule → code)
4. Is the action at the output reversible? (irreversible → gate)
5. How to verify the fact? (class of error + principle)
6. Is the decision regulated? (reason codes / appeal)
7. Who is liable and where is a human mandatory?
8. Is there PII / financial data / biometrics? → not into an LLM; 152-FZ

Assignment — Seminar 5 «Fact-checking AI on financial data»

In ~30 min, teams verify 5 real claims against primary sources themselves (Bank of Russia / VCIOM), marking them true/partial/false. The lecture teaches recognizing the class (Understand); the seminar — verifying independently (Apply).

Questions

The palette of types — a lens for all the industry lectures ahead. Thank you for your attention.